AURA V1 — C

Cross-check: "All component.docx" v

S.No	Component	In Docx?	In V1 Specs?	Status
1	ESP32 (WROOM DevKit)	Yes	Yes — Required	OK
2	3.5-inch Touchscreen (TFT Display)	Yes	Yes — Required	OK
3	DHT22 Temperature & Humidity Sensor	Yes	Yes — Required	OK
4	DS3231 Real-Time Clock (RTC)	Yes	Yes — Required	OK
5	DFPlayer Mini (MP3-TF-16P V3.0)	Yes	Yes — Required	WRONG LINK
6	Speaker	Yes	Yes — Required	OK
7	INMP441 Microphone	Yes	Yes — Required	OK
8	Camera (ESP32-CAM)	Yes	Yes — Required	OK
9	MicroSD Card Reader	Yes	Yes — Required	OK
10	5V Buzzer	Yes	Yes — In Pinmap	OK
11	PAM8403 Amplifier	Yes	Yes — In Pinmap	OK
12	Buck Converter (LM2596)	Yes	Yes — In Pinmap	OK
13	PIR Motion Sensor	NO — MISSING	Yes — Required	MISSING
14	Touch Input / Controller	NO — MISSING	Yes — Required	MISSING
15	AQI Sensor (MQ-135)	Yes	NO — Not in V1 Specs	EXTRA
16	Charging Module (Multi-Cell Li Charger + Boost)	Yes	Partially — Power deferred	WARNING
17	Battery Cells (4× 18650)	Yes	Partially — 1S documented	WARNING

Component Verification Report

s AURA_V1_REQUIREMENTS.md + AURA_PINMAP_V1.md

Issue / Notes	Priority
Present in docx as item #1. Matches V1 requirement. Main controller for AURA.	High
Present in docx as item #2. Matches V1 requirement. Pin assignment still TBD per pinmap.	High
Present in docx as item #3. Matches V1 requirement. Assigned to GPIO 27.	High
Present in docx as item #4. Matches V1 requirement. I²C on GPIO 21/22.	High
Present in docx as item #5, BUT the purchase link points to the 3.5" TFT screen page (quartzcomponents touchscreen URL), NOT to a DFPlayer Mini product. The link is INCORRECT.	High
Present in docx as item #6. Matches V1 requirement. Pinmap shows 6Ω/15W speaker via PAM8403 amp.	High
Present in docx as item #7. Matches V1 requirement. I²S on GPIO 26/25/33.	High
Present in docx as item #8. Matches V1 requirement. Separate vision subsystem board.	High
Present in docx as item #12. Matches V1 requirement. SPI CS on GPIO 5.	Medium
Present in docx as item #13. In AURA_PINMAP_V1 on GPIO 13. Not in V1 requirements table but is in the architecture.	Medium
Present in docx as item #15. In AURA_PINMAP_V1 audio chain (DFPlayer → PAM8403 → Speaker). Essential for audio output.	High
Present in docx as item #14. Referenced in AURA_PINMAP_V1 Section 14. Pending Day 5 power audit verification.	Medium
NOT in docx component list! V1 Requirements table lists PIR as Required (Medium priority). Pinmap reserves GPIO 34 for it. You need to add this component to your list.	Medium
NOT listed as a separate component in docx. V1 Requirements table lists Touch Input as Required (High priority). The touchscreen TFT should include a touch controller (likely XPT2046), but it is not explicitly listed or verified.	High
Present in docx as item #10, but NOT listed in AURA_V1_REQUIREMENTS.md or AURA_PINMAP_V1.md. This is OUTSIDE the frozen V1 scope. Adding it would violate scope rules (Section 8 of requirements).	N/A
Present in docx as item #9. Pinmap mentions battery/charging but architecture is NOT frozen. Docx says 'Multi-Cell' charger with 4 cells, but pinmap records only 1× 18650 (1S). Major mismatch with current 1S architecture.	Medium
Docx lists 4× 18650 cells (item #11), but AURA_PINMAP_V1 documents only 1× 18650 3.7V 3000mAh (1S). The earlier project had a 2S concept. 4-cell arrangement is NOT in V1 frozen specs.	Medium

Action Required
None — verified
None — verified
None — verified
None — verified
Fix the purchase link to point to the actual DFPlayer Mini product page
None — verified
None — verified
None — verified
None — verified
None — verified
None — verified
None — verified
ADD to component list. Purchase a PIR motion sensor (e.g., HC-SR501).
Verify that the ordered 3.5" TFT includes an integrated touch controller. If not, purchase separately.
REMOVE from V1 scope or get explicit approval to add. Do NOT integrate during 90-day roadmap.
Reconcile: is AURA using 1S (1 cell) or 4S (4 cells)? Day 5 power audit must resolve this before purchase.
Reconcile battery architecture. 1S vs 2S vs 4S must be decided in Day 5 power audit.

Verification Summary

Total Components (Docx)	15
Total Required (V1 Specs)	11
Verified OK	9
Wrong Link	1
Missing from Docx	2
Extra (Not in V1 Scope)	1
Warning (Need Clarification)	2

#	Component	Issue Type
1	DFPlayer Mini	WRONG LINK
2	PIR Motion Sensor	MISSING
3	Touch Input/Controller	MISSING
4	AQI Sensor (MQ-135)	EXTRA — OUT OF SCOPE
5	Battery: 4× 18650 vs 1× 18650	MISMATCH
6	Charging Module: Multi-Cell vs 1S	MISMATCH

Details & Fix
Purchase link in docx points to 3.5" TFT screen URL, NOT DFPlayer Mini. Fix the link to point to actual DFPlayer Mini product.
Not listed in docx at all. V1 Requirements lists it as Required (Medium). Reserved on GPIO 34. Add HC-SR501 or equivalent PIR sensor to your component list.
Not listed separately in docx. V1 Requirements lists Touch Input as Required (High). Verify if your ordered 3.5" TFT includes integrated touch (XPT2046). If standalone, add it.
Listed in docx but NOT in V1 frozen specs. Per Scope Rules Section 8: 'New features should not be added without reviewing their effect on the roadmap.' Remove from V1 or get explicit approval.
Docx lists 4-cell arrangement. Pinmap documents 1S single cell. Previous project had 2S concept. Must reconcile in Day 5 power audit before purchasing.
Docx lists a Multi-Cell charger + boost, but V1 architecture is 1S. If you use 1S, you need a different (simpler) TP4056-based charger. Resolve in Day 5.